

Competency

- To solve the daily life problems by using the basic rules of household arithmetic and financial arithmetic.

Learning Outcomes

- To solve the problems on household arithmetic such as electricity bills, telephone bills, water bills, taxi meter reading (including discount and VAT)

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define 1 unit of electricity - To tell the formula of finding the consumed units - To tell the formula of finding the tariff (Sub-total) - To write the relation of total charge, tariff, TSC and VAT - To define 1 unit of water consumption - To name the concerned authority of ministry of water supply - To name the concerned authority for implement the rules and regulations regarding taximeter. - To recall the minimum taxi fares during 6:00 am to 9:00 pm or 9:00 pm to 6:00 am.
2.	Understanding (U)	- To find the electricity charge for simple problems - To calculate the telephone charge - To calculate the taxi fare
3.	Application (A)	- To find the electricity charge with rebate/fine - To find the number of consumed units of electricity - To apply the rules and find the water charge - To find the telephone charge with TSC and VAT
4.	High Ability (HA)	- To estimate the consumed unit of electricity of a month and find the bill amount as per present rate of electricity - To prepare report on few ways of reducing unnecessary use of electricity/telephone/water and use of ICT used for payment of the bills

Required Teaching Materials/ Resources

Electricity bills, models of meter box, telephone bills, water bills, calculator, recharge card, chart paper mentioning the taxi fare rates etc.

Pre-knowledge: discount, VAT

A. Electricity bills

Teaching Activities

1. Ask about the electric appliances used in the students' houses.
2. With some samples/models of meter readings of two successive months, ask the following questions:
 - (i) What is the reading of recent month?
 - (ii) What is the reading of preceding month?
 - (iii) What is the number of consumed units of electricity?
 - (iv) What do you mean by 1 unit of consumed electricity?
3. Present the recent rates, rules and regulations of electricity fixed by Nepal Electricity Authority (NEA) on chart paper or power point presentation or through available website.
4. Encourage the students to find the electricity charge with electricity service charge and rebate/fine.

Solution of selected problems from Vedanta Excel in Mathematics

1. **The rate of electricity charge up to 20 units is Rs 3 per unit and Rs 7 per unit from 21 to 30 units. Find the charge of consumption of 28 units with Rs 50 service charge.**

Solution:

Here, consumption of electricity = 28 units

Rate of charge up to 20 units = Rs 3 per unit

∴ Charge up to 20 units = $20 \times \text{Rs } 3 = \text{Rs } 60$

The excessive number of units = $28 - 20 = 8$ units

∴ Charge of excessive 8 units = $8 \times \text{Rs } 7 = \text{Rs } 56$

Total charge of electricity with service charge = $\text{Rs } 60 + \text{Rs } 56 + \text{Rs } 50 = \text{Rs } 166$

2. **Electricity tariff rates and rebate/fine rules are given below**

kWh (Monthly) Units	5 Ampere		15 Ampere		30 Ampere		60 Ampere	
	Service Charge	Energy Charge per unit	Service Charge	Energy Charge per unit	Service Charge	Energy Charge per unit	Service Charge	Energy Charge per unit
0 – 20	Rs30.00	Rs 3.00	Rs50.00	Rs4.00	Rs75.00	Rs5.00	Rs125.00	Rs6.00

The rules of rebate/fine

Meter reading	Within 7 days	Within 8-22 days	Within 23-30 days	Within 31-40 days	Within 41-60 days
Rebate/fine	3% rebate	-	5% fine	10% fine	25% fine

From the tables given above, workout the following problems

- a) **A household having 5A electricity meter consumed 18 units of electricity in one month. Find the amount of payment made by the household within 7 days.**
- b) **A household having a 15A meter consumed 16 units of electricity in one month. Find the amount of payment made by the household on 25th day of meter reading.**
- c) **A household having a 30A meter consumed 19 units of electricity in one month and if the payment was made on 20th day of meter reading, find the amount of payment.**

- d) A household having a 60A meter consumed 20 units of electricity in one month and the payment was made on 50th day of meter reading. Calculate the amount of payment made by the house.

Solution:

- (a) Here, capacity of meter box = 5A, consumption of electricity = 18 units
Rate of charge up to 20 units = Rs 3 per unit
Now, charge for 18 units = $18 \times \text{Rs } 3 = \text{Rs } 54$
 $\therefore$ Total charge of electricity with service charge = Rs 54 + Rs 30 = Rs 84
Since, the payment was made within 7 days of meter reading. So, 3% rebate was allowed.
Hence, the required payment was Rs 84 – 3% of Rs 84 = Rs 81.48
- (b) Here, capacity of meter box = 15A, consumption of electricity = 16 units
Rate of charge up to 20 units = Rs 4 per unit
Now, charge for 16 units = $16 \times \text{Rs } 4 = \text{Rs } 64$
 $\therefore$ Total charge of electricity with service charge = Rs 64 + Rs 50 = Rs 114
Since, the payment was made on 25th day of meter reading. So, 5% fine was charged.
Hence, the required payment was Rs 114 + 5% of Rs 114 = Rs 119.70
- (c) Here, capacity of meter box = 30A, consumption of electricity = 19 units
Rate of charge up to 20 units = Rs 5 per unit
Now, charge for 19 units = $19 \times \text{Rs } 5 = \text{Rs } 95$
 $\therefore$ Total charge of electricity with service charge = Rs 95 + Rs 75 = Rs 170
Since, the payment was made on 20th day of meter reading. So, there was no rebate no fine.
Hence, the required payment was Rs 170.
- (d) Here, capacity of meter box = 60A, consumption of electricity = 20 units
Rate of charge up to 20 units = Rs 6 per unit
Now, charge for 20 units = $20 \times \text{Rs } 6 = \text{Rs } 120$
 $\therefore$ Total charge of electricity with service charge = Rs 120 + Rs 125 = Rs 245
Since, the payment was made on 50th day of meter reading. So, 25% fine was charged.
Hence, the required payment was Rs 245 + 25% of Rs 245 = Rs 306.25
3. Mr. Sharma has a 5A meter in his house. He uses 5 CFL bulbs of 15 watt each for 4 hours and an electric heater of 1200 watt for 1 hour every day. Find the cost of payment of the bill of the month at the rate of Rs 3 per unit up to 20 units, Rs 7 per unit from 21 to 30 units and Rs 8.50 from 31-50 units with Rs 75 service charge, if the payment is made on 10th day of the meter reading.

Solution:

Here, consumption of electricity in 1 day = $(5 \times 15 \times 4 + 1200 \times 1)$ watts = 1500 watts

Consumption of electricity in 1 month = 30×1500 watts = 45000 watts = 45 kW

$\therefore$ Number of consumed units = 45 units because the time duration of use of electric appliances were measured in hours.

Now,

Consumption block	No. of units	Rate of charge	Electricity charge
0 – 20	$20 - 0 = 20$	Rs 3	$20 \times \text{Rs } 3 = \text{Rs } 60$
21 – 30	$30 - 20 = 10$	Rs 7	$10 \times \text{Rs } 7 = \text{Rs } 70$
31 – 50	$45 - 30 = 15$	Rs 8.50	$15 \times \text{Rs } 8.50 = \text{Rs } 127.50$

Total charge of electricity with service charge = energy charge + service charge
 = Rs 60 + Rs70 + Rs 127.50 + Rs 75 = Rs 332.50

Since, the payment was made on 10th day of meter reading. So, there was neither rebate nor fine.

Hence, the required payment was Rs332.50

4. **Mrs. Bajracharya's house has a 15A meter. She uses 5 LED bulbs of 10 watt each for 4 hours, 2 televisions of 60 watt each for 5 hours and a refrigerator of 250 watt for 2 hours every day. Find the cost of payment of the bill of the month at the rate of Rs4 per unit up to 20 units, Rs 7 per unit from 21 to 30 units and Rs 8.50 from 31-50 units with Rs100 service charge, if the payment is made on 35th day of the meter reading.**

Solution:

Here, consumption of electricity in 1 day = $(5 \times 10 \times 4 + 2 \times 60 \times 5 + 1 \times 250 \times 2)$ watts
 = 1300 watts

Consumption of electricity in 1 months = 30×1300 watts = 39000 watts = 39 kW

∴ Number of consumed units = 39 units because the time duration of use of electric appliances were measured in hours.

Now,

Consumption block	No. of units	Rate of charge	Electricity charge
0 – 20	$20 - 0 = 20$	Rs4	$20 \times \text{Rs}4 = \text{Rs } 80$
21 – 30	$30 - 20 = 10$	Rs 7	$10 \times \text{Rs } 7 = \text{Rs } 70$
31 – 50	$39 - 30 = 9$	Rs 8.50	$9 \times \text{Rs } 8.50 = \text{Rs}76.50$

Total charge of electricity with service charge = energy charge + service charge
 = Rs 80 + Rs 70 + Rs76.50 + Rs100 = Rs 326.50

Since, the payment was made on 35th day of meter reading. So, 10% fine was added

Hence, the required payment was Rs326.50 + 10% of Rs 326.50 = Rs 359.15

5. **The meter box of a family house is 15 A. If the family made the payment of Rs 1336.50 with service charge of Rs125 on 36th day of meter reading, how many units of electricity was consumed in the month? Calculate it under the following rates.**

Units	0 – 20	21 – 30	31 – 50	51 – 150
Rate of charge per unit	Rs 4	Rs 7	Rs 8.50	Rs 11

Payment up to 40th day from the meter reading – 10% fine

Solution:

Let the number of consumed units in the month be x units

Now,

Consumption block	No. of units	Rate of charge	Electricity charge
0 – 20	$20 - 0 = 20$	Rs 4	$20 \times \text{Rs } 4 = \text{Rs } 80$
21 – 30	$30 - 20 = 10$	Rs 7	$10 \times \text{Rs } 7 = \text{Rs } 70$
31 – 50	$50 - 30 = 20$	Rs 8.50	$20 \times \text{Rs } 8.50 = \text{Rs}170$
51 – 150	$x - 50$	Rs 11	$(x - 50) \times \text{Rs } 11 = \text{Rs } (11x - 550)$

Total charge of electricity with service charge = energy charge + service charge

$$= \text{Rs } 80 + \text{Rs } 70 + \text{Rs } 170 + 11x - \text{Rs } 550 + \text{Rs } 125 = \text{Rs}(11x - 105)$$

Also, extra fine = 10% of Rs $(11x - 105) = \text{Rs } (1.1x - 10.5)$

According to question,

Payment of bill with fine = Rs 1336.50

$$\text{or, Rs } (11x - 105) + \text{Rs } (1.1x - 10.5) = \text{Rs } 1336.50 \quad \therefore x = 120$$

Hence, the required payment number of consumed units is 120.

Extra Questions

- The rate of electricity charge up to 20 units is Rs 3 per unit and Rs 7 per unit from 21 to 30 units. Find the charge of consumption of 24 units with Rs 50 service charge.
[Ans: Rs138]
- The rate of electricity charge up to 20 units is Rs 3 per unit and Rs 7 per unit from 21 to 30 units. If a family paid the bill of Rs 166 with Rs 50 service charge, how many units of electricity was consumed in the month?
[Ans: 28]
- The meter readings of Krishna's house in 1st Asar was 02967 and 1st Sharwan was 03015. Find the electricity charge for the month of Asar according to the given information if the bill was made on 5th day from meter reading.

KWh (units)	Service charge	Energy charge per unit
0 – 20	Rs 30	Rs 3
21 – 30	Rs 50	Rs 7
31 – 50	Rs 75	Rs 8.50

Payment within 7 days of meter reading – 3% rebate

[Ans: Rs347.26]

- The meter box of a family house is 15 A. If the family made the payment of Rs1331 with service charge of Rs125 on 36th day of meter reading, how many units of electricity was consumed in the month? Calculate it under the following rates.

Units	0 – 20	21 – 30	31 – 50	51 – 150
Rate of charge per unit	Rs 4	Rs 7	Rs 8.50	Rs 11

Payment up to 40th day from the meter reading – 10% fine

[Ans: 110 units]

B. Telephone bills

Teaching Activities

- With a recharge card, discuss about the mobile network service
- Provide some telephone bills to the students, and ask the following questions:
 - What is concerned authority for the implementation of this bill?
 - To whom is the telephone bill issued?
 - What is the address of the telephone line?
 - What type of telephone is it?
 - What status of telephone is mentioned?
 - What type of telephone is it?
 - What is the previous reading of telephone given in the bill?
 - What is the current reading of telephone given in the bill?
 - How many telephone calls is made in the month?
 - What is the rental amount?
 - How many extra calls are made in the month?

- (xii) What is the sub-total amount given in the bill?
 - (xiii) What amount is to be paid for TSC?
 - (xiv) What amount of VAT is to be paid?
 - (xv) What is the grand total given in the bill?
3. Explain the telephone billing system implemented by Nepal Doorsanchar Company Ltd.
 4. List the following formulae after discussion
 - (i) $\text{Tariff} = \text{Sub-total} = \text{Minimum charge} + \text{Extra charge}$
 - (ii) $\text{Telecom service charge (TSC)} = 10\% \text{ of Sub-total}$
 - (iii) $\text{Total} = \text{Sub-total} + \text{TSC}$ (iv) $\text{VAT amount} = \text{VAT\% of Total} = 13\% \text{ of Sub-total}$
 - (v) $\text{Grand total} = \text{Total} + \text{VAT amount}$

Solution of selected problems from Vedanta Excel in Mathematics

1. ***The minimum charge up to 175 calls is Rs 200. If the charge for each additional call is Re 1, how much will be the charge for 475 calls with 10% TSC and 13% VAT?***

Solution:

Here, minimum charge up to 175 calls = Rs 200

The additional number of calls = $475 - 175 = 300$

The charge for additional calls = $300 \times \text{Re } 1 = \text{Rs } 300$

Now, sub-total = Minimum charge + additional charge = $\text{Rs } 200 + \text{Rs } 300 = \text{Rs } 500$

TSC = 10% of Rs 500 = Rs 50

Also, total = sub-total + TSC = $\text{Rs } 500 + \text{Rs } 50 = \text{Rs } 550$

Again, grand-total = Total + VAT% of Total = $\text{Rs } 550 + 13\% \text{ of Rs } 550 = \text{Rs } 621.50$

Hence, the total charge for 375 calls is Rs 621.50

2. ***The minimum charge of telephone calls up to 175 calls is Rs 200. The charge for each extra call of 2 minutes duration is Re 1. If the household paid Rs 633.93 with 10% TSC and 13% VAT to clear the bill of the month, find the total number of calls made in the month?***

Solution:

Let the charge of telephone calls without VAT be Rs x.

Then, grand total = Total + VAT% of total

or, $\text{Rs } 633.93 = x + 13\% \text{ of } x \therefore x = \text{Rs } 561$

Also, let the charge of telephone calls without TSC be Rs y.

Then, total = sub-total + TSC or, $\text{Rs } 561 = y + 10\% \text{ of } y \therefore y = \text{Rs } 510$

Again, the charge for extra calls = $\text{Rs } 510 - \text{Rs } 200 = \text{Rs } 310$

Now, the number of extra calls = $\frac{\text{Rs } 310}{\text{Re } 1} = 310$

Hence, the total number of calls of the month is $175 + 310 = 485$ calls

Extra Questions

1. The minimum charge up to 175 calls is Rs 200. If the charge for each additional call is Re 1, how much will be the charge for 375 calls with 10% TSC and 13% VAT?
[Ans: Rs 497.20]
2. The reading of the Baishakh-1 of local calls in Supriya's house is 5270 and that of the Jestha-1 is 5605. The minimum charge up to 175 calls is Rs 200 and charge for each additional call is Re 1.

- (i) How many calls are made in the month of Baishakh?
(ii) What is the bill amount of telephone calls with 10% TSC and 13% VAT?

[Ans: (i) 635 calls, (ii) Rs 820.38]

3. The minimum charge of telephone calls up to 175 calls is Rs 200. The charge for each extra call of 2 minutes duration is Re 1. If the household paid Rs 559.35 with 10% TSC and 13% VAT to clear the bill of the month, find the total number of calls made in the month?

[Ans: 425 calls]

C. Water bills

Teaching Activities

- Provide some water bills to the students, and ask the following questions:
 - What is concerned authority for the implementation of this bill?
 - How many units of water is consumed?
 - What is the size of pipe?
 - What is the previous reading of water consumption?
 - What is the present reading of water consumption?
 - What is the minimum charge?
 - What is the additional charge?
 - What is the total charge?
- Discuss about 1 unit of water consumption
- Show the water tariff rules in the chart paper implemented by Nepal Water Supply Corporation.

S.N.	Size of pipe	Tap with meter			Taps without meter	
		Minimum consumption (litre)	Minimum Charge	Additional consumption per 1000 litre (Rs)	Main Tap Charge(Rs)	Branch Tap Charge (Rs)
1.	$\frac{1}{2}$ "	10,000	110	25	560	200
2.	$\frac{3}{4}$ "	27, 000	1490	40	3360	1600
3.	1"	56,000	3420	40	9200	2700

The compulsory provision of sewerage service charge = 50% of the water consumption charge.

Payment schedule of the bill

S.N.	Payment is made after the bill issued	Rebate/ Fine
1.	Within the 1 st and the 2 nd month	3% rebate
2.	Within the 3 rd month	No rebate and no fine
3.	Within the 4 th month	10% fine
4.	Within the 5 th month	20% fine
5.	After 5 th month	50% fine

4. Present the water tariff rules implemented by Kathmandu Upatyaka Khanepani Limited (KUKL) in the chart paper.

S.N.	Size of pipe	Tap with meter			Taps without meter
		Minimum consumption (litre)	Minimum Charge	Additional consumption per 1000 litre (Rs)	Branch Tap Charge (Rs)
1.	$\frac{1}{2}$ "	10,000	100	32	785
2.	$\frac{3}{4}$ "	27, 000	1910	71	4595
3.	1"	56,000	3960	71	9540

Solution of selected problems from Vedanta Excel in Mathematics

1. 127 units of water is consumed by using $\frac{3}{4}$ " pipe in a hotel. If the payment of the bill is made within the fifth month after the bill issued, how money is required to clear the bill with 50% sewerage service charge?

Solution:

Here,

According to the water tariff provisions of NWS for the use of pipe of the size $\frac{3}{4}$ "

The minimum charge up to 27000 litres i.e., 27 units = Rs1490

The charge of additional units = Rs 40 per unit

The additional number of units = $127 - 27 = 100$ units

The charge for additional units = $100 \times \text{Rs } 40 = \text{Rs } 4000$

Now, total charge = Rs 1490 + Rs 4000 = Rs 5,490

Again, the charge including 50% sewerage service = Rs 5490 + 50% of Rs 5490 = Rs 8235

2. A household uses $\frac{1}{2}$ " of water pipe. The meter reading of the household on 1st of Asar was 1260 units and on 1st of Shrawan was 1330 units. Calculate the charge to be paid including 50% sewerage service charge if the payment of the bill is made in the following schedule.

- Within the first month after the bill issued
- Within the third month after the bill issued
- Within the fifth month after the bill issued
- Within the seventh month after the bill issued

Solution:

Here,

The meter reading of 1st Asar = 1260 and that of 1st Shrawan = 1330

$\therefore$ Consumed units of water = $1330 - 1260 = 70$

According to the water tariff provisions of NWS for the use of pipe of the size $\frac{1}{2}$ "

The minimum charge up to 10000 litres i.e., 10 units = Rs110

The charge of additional units = Rs 25 per unit

The additional number of units = $70 - 10 = 60$ units

The charge for additional units = $60 \times \text{Rs } 25 = \text{Rs } 1500$

Now,

Total charge = Rs110 + Rs 1500 = Rs 1610

Again, the charge including 50% sewerage service = Rs 1610 + 50% of Rs 1610 = Rs 2415

(i) When the payment of the bill is made within the first month after the bill issued, 3% rebate is allowed

∴ Required payment = Rs 2415 – 3% of Rs 2415 = Rs 2342.55

(ii) When the payment of the bill is made within the third month after the bill issued, there is no rebate no fine.

∴ Required payment = Rs 2415

(iii) When the payment of the bill is made within the fifth month after the bill issued 20% fine is charged

∴ Required payment = Rs 2415 + 20% of Rs 2415 = Rs 2898

(iv) When the payment of the bill is made within the seventh month after the bill issued, 50% fine is charged

∴ Required payment = Rs 2415 + 50% of Rs 2415 = 3622.50

Extra Questions

- 18 units of water is consumed by using $\frac{1}{2}$ " pipe in Rameshwor's house. If the payment of the bill is made within the second month after the bill issued, how money is required to clear the bill with 50% sewerage service charge? [Ans:Rs 451.05]
- 147 units of water is consumed by using $\frac{3}{4}$ inch pipe in Everest Hotel. If the payment of the bill is made within the fourth month after the bill issued, how much amount is required to pay as bill? [Ans: Rs 10,378.50]
- A household uses $\frac{1}{2}$ " of water pipe. The meter reading of the household on 1st of Kartik was 1420 units and on 1st of Mansir was 1480 units. Calculate the charge to be paid including 50% sewerage service charge if the payment of the bill is made in the following schedule.
 - Within the second month after the bill issued
 - Within the third month after the bill issued
 - Within the fifth month after the bill issued
 - Within the sixth month after the bill issued

[Ans: (i) Rs 1076.70, (ii) Rs 1110, (iii) 1332 (iv) 1665]

D. Taxifare

Teaching Activities

- Discuss upon taxi fare paid by parents or students themselves
- Present the rules and regulations implemented and monitored by Nepal Bureau of Standards and Metrology (NBSM) regarding the taxi fare as shown in the following table.

Time	Minimum fare	Fare of per 200 meters	Waiting charge per 2 minutes
6:00 am to 9:00 pm	Rs 14	Rs 7.20	Rs 7.20
9:00 pm to 6:00 am	Rs 21	Rs 10.80	Rs 10.80

Solution of selected problems from Vedanta Excel in Mathematics

1. **Mr. Kattel travelled 15 km by a hired taxi at 5:00 am. The minimum fare of Rs 21 appeared immediately after the meter was flagged down. Then, the fare went on at the rate of Rs 10.80 per 200 metres. An additional waiting charge of Rs 10.80 per 2 minutes was charged for waiting of 10 minutes. Calculate the total of the taxi fare paid by him.**

Solution:

Here, the minimum fare = Rs 21, distance travelled = 15 km = 15000 m

Now,

The fare of 200 metres = Rs 10.80 or, the fare of 1 m = Rs 0.054

∴ The fare of 15000 m = $15000 \times \text{Rs } 0.054 = \text{Rs } 810$

Also,

Waiting charge of 2 minutes = Rs 10.80 ∴ Waiting charge of 10 minutes = Rs 54

Hence, the total fare = Rs 21 + 810 + Rs 54 = Rs 885

2. **Rita hired taxi and travelled a certain distance at 8:00 a.m. She paid the total fare of Rs 194. If the minimum fare is Rs 14 and the fare per 200 metres is Rs 7.20, find the distance travelled by her.**

Solution:

Here, minimum charge = Rs 14

∴ The taxi fare excluding minimum fare = Rs 194 – 14 = Rs 180

Now, Rs 7.20 is the fare of 200 metres or, Re 1 is the fare of $\frac{200}{7.20}$ metres

∴ Rs 180 is the fare of $\frac{200}{7.20} \times 180 = 5000$ meters = 5 km

Hence, she travelled 5 km.

Extra Questions

1. Mr. Koirala travelled 7 km by a hired taxi at 1:30 p.m. The minimum fare of Rs 14 appeared immediately after the meter was flagged down. Then, the fare went on at the rate of Rs 7.20 per 200 metres. Calculate the total of the taxi fare paid by him.
[Ans: Rs 266]
2. Mrs. Maharjan travelled 10 km by a hired taxi at 4:45 am. The minimum fare of Rs 21 appeared immediately after the meter was flagged down. Then, the fare went on at the rate of Rs 10.80 per 200 metres. An additional waiting charge of Rs 10.80 per 2 minutes was charged for waiting of 10 minutes. Calculate the total of the taxi fare paid by him.
[Ans: Rs 615]
3. Smriti hired taxi and travelled a certain distance at 7:15 a.m. She paid the total fare of Rs 158. If the minimum fare is Rs 14 and the fare per 200 metres is Rs 7.20, find the distance travelled by her.
[Ans: 4 km]
4. Rupesh hired taxi and travelled a certain distance at 10:00 p.m. He paid the total fare of Rs 1155 including the waiting charge of 10 minutes. Find the distance travelled by him.
[Ans: 4 km]